Computational Mathematics

Sheet II.1 — HT25

1. Prove by induction that if Euclid's algorithm requires N divisions to compute the greatest common divisor of the pair $a > b, a, b \in \mathbb{N}$, then $a \geq F_{N+2}$ and $b \geq F_{N+1}$, where F_i denotes the ith Fibonacci number ($F_1 = 1, F_2 = 1, F_{i+2} = F_{i+1} + F_i$).

2. (a) Given $a, b, c \in \mathbb{Z}$, implement the extended Euclidean algorithm to find a solution (x, y) (if any exists) of

$$ax + by = c.$$

Use your code to find a solution to $993319x + 999331y = 939193$.

(b) The Rivest–Shamir–Adleman (RSA) cryptographic system is one of the most important protocols for securely exchanging messages. In most cryptographic systems, the key used for encryption and decryption are the same; in RSA, the key used for encryption is public, but the key used for decryption is kept secret.

The public key consists of two natural numbers, n and e. Here $n = pq$ where p and q are primes; p and q are kept secret. Knowledge of p and q lets us compute the *Carmichael totient function*

$$\lambda(n) = \frac{(p-1)(q-1)}{\gcd(p-1, q-1)}.$$

The public key must satisfy $1 < e < \lambda(n)$ and $\gcd(e, \lambda(n)) = 1$.

The associated private key d satisfies

$$de \equiv 1 \mod \lambda(n).$$

Using your implementation in part (a), compute the private key d for the public key $n = 137 \cdot 173, e = 7$.

(c) *(Optional)* encrypt the message $m = 21051$ using the formula

$$c \equiv m^e \mod n$$

and decrypt the resulting ciphertext c using

$$m \equiv c^d \mod n.$$

Patrick Farrell: patrick.farrell@maths@maths.ox.ac.uk

1. Prove by induction that if Euclid's algorithm requires N divisions to compute the greatest common divisor of the pair $a > b, a, b \in \mathbb{N}$, then $a \geq F_{N+2}$ and $b \geq F_{N+1}$, where F_i denotes the ith Fibonacci number ($F_1 = 1, F_2 = 1, F_{i+2} = F_{i+1} + F_i$).

induct on big N?

Suppose 1 divisions is needed

then $a \geq 2 = F_{2+1}$ and $b \geq 1 = F_{1+1}$ so true in BC.

Suppose that the result holds for some $N = k$

So if k divisions are needed then

$a \geq F_{k+2}$, $b \geq F_{k+1}$

Consider (c, d), for which $k+1$ divisions are needed

Then $c = kd + \mu$

Since $k+1$ divisions are needed, $\mu \neq 0$

$\Rightarrow (d, \mu)$ requires k divisions

$\Rightarrow d \geq F$ $\quad \mu \geq F$

$\Rightarrow \quad \alpha \geq F_{k+2}, \quad \mu \geq F_{k+1}$

$\lambda \geq 1$

$\Rightarrow \quad c \geq F_{k+2} + F_{k+1}$

$\geq F_{k+3}$

As needed

(b) The Rivest–Shamir–Adleman (RSA) cryptographic system is one of the most important protocols for securely exchanging messages. In most cryptographic systems, the key used for encryption and decryption are the same; in RSA, the key used for encryption is public, but the key used for decryption is kept secret.

The public key consists of two natural numbers, n and e. Here $n = pq$ where p and q are primes; p and q are kept secret. Knowledge of p and q lets us compute the *Carmichael totient function*

$$\lambda(n) = \frac{(p-1)(q-1)}{\gcd(p-1, q-1)}.$$

The public key must satisfy $1 < e < \lambda(n)$ and $\gcd(e, \lambda(n)) = 1$.

The associated private key d satisfies

$$de \equiv 1 \mod \lambda(n).$$

Using your implementation in part (a), compute the private key d for the public key $n = 137 \cdot 173, e = 7$.

$p = 137, \quad q = 173$

$\Rightarrow$

$7d \equiv 1 \pmod{\lambda n}$

$\Rightarrow \quad 7d = 1 - \mu\lambda$

$7d + \lambda\mu = 1$

3. Classify the order of convergence of the following sequences:
   (a) $a_n = 2^{-n}$.
   (b) $a_n = (n+1)^{-1}$.
   (c) $a_n = 2^{-2^n}$.

   For each sequence, plot a_n from $n = 1$ up to N, where $N = 100$ for (a) and (b) and $N = 10$ for (c), using a logarithmic scale on the y-axis.

4. Does Banach's contraction mapping theorem apply to the following continuous real-valued functions g with domain of definition X? Justify your answers.
   (a) $X = \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right], \quad g(x) = \sin x + x$.
   (b) $X = [0, 2], \quad g(x) = x^3 + 3x^2 - 3$.
   (c) $X = [1, \infty), \quad g(x) = x + x^{-1}$.

5. (a) Let $a_n \to a$ be a convergent sequence of real numbers. Let $e_n := |a_n - a|$ be its associated sequence of errors. Assume that the error can be described as

$$e_{n+1} \approx M e_n^q$$

   for constants M and q.

   Given three successive terms e_n, e_{n+1}, e_{n+2} in the sequence of errors, describe a procedure to estimate M and q.

   (b) Consider the Leibniz formula for π:

$$\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1},$$

   and let a_n be its sequence of partial sums.

   Estimate M and q for this sequence, starting from e_n with $n = 10,000$, and thereby state a conjecture about its convergence order.

   (c) Recall that the Aitken acceleration $\tilde{a}_n$ of a sequence a_n is given by

$$\tilde{a}_n = a_n - \frac{(a_{n+1} - a_n)^2}{a_{n+2} - 2a_{n+1} + a_n}.$$

   Plot the sequence of errors up to $n = 10,000$ of the partial sums a_n of the Leibniz formula for π, as well as its Aitken acceleration $\tilde{a}_n$, and the iterated Aitken acceleration $\tilde{\tilde{a}}_n$.

   (d) What is the estimated order of convergence of $\tilde{a}_n$ and $\tilde{\tilde{a}}_n$? Comment on your observations.

   [*Hint: you may wish to use higher-precision arithmetic for this question. Use* `mpmath` *with* `mpmath.mp.dps = 100`.]

Patrick Farrell: patrick.farrell@maths@maths.ox.ac.uk